

PMF (Probability Mass Function)

PMF is referred to as the probability of discrete random variable which is equal to a particular value. It is represented as $f(x) = P(X = x)$

where

- X is discrete random variable
- x is the specified value.

Example:

Let is dice is rolled then the probability of getting a number equal to 4 is an example of probability mass function. The sample space for the given event is $\{1, 2, 3, 4, 5, 6\}$ and X be the random variable for getting a 4. The probability mass function evaluated for $X = 4$ is $1/6$.

Probability Mass Function Formula in Binomial Distribution

[Binomial distribution](#) with the number of outcomes, probability of success and probability of failures. The PMF formula in Binomial distribution is given by:

$$P(X = x) = {}^nC_x p^x (1 - p)^{n-x}$$

where,

- n is Number of Outcomes
- p is Probability of Success
- (1-p) is Probability of Failure

Probability Mass Function Formula in Poisson Distribution

[Poisson distribution](#) deals with the mean and the number of independent events occurred in specific interval of time. The formula for the probability mass function in Poisson distribution is given by:

$$P(X = x) = [\lambda^x e^{-\lambda}] / x!$$

where,

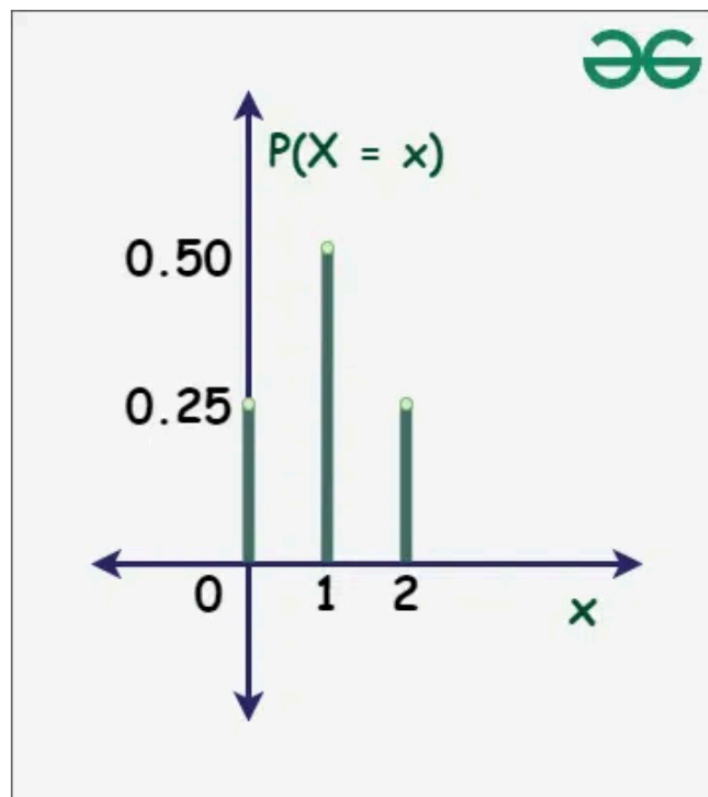
- λ is [Mean](#)

Probability Mass Function Table and Graph

Table that represents the probability mass function with the value of the random variables is called the probability mass function. Let a coin is tossed two times and X be the random variable representing the number of tails then, the probability mass function table for above event is given below.

x	0{HH}	1{HT, TH}	2{TT}
$P(X = x)$	$1/4$	$2/4 = 1/2$	$1/4$

Probability mass function graph for above table is given below:



Probability Mass Function

Properties of Probability Mass Function

Some properties of the probability mass function are listed below.

- $f(x) = P(X = x) > 0$ hence, probability is always positive.
- $\sum_{x \in S} f(x) = 1$ hence, sum of all probabilities equals to 1.
- $P(X \in E) = \sum_{x \in E} f(x)$ hence, probability of event E is given by sum of probabilities of values of x in E. It is used to determine CDF.

PROBABILITY MASS FUNCTION	PROBABILITY DENSITY FUNCTION
The PMF is the probability that a discrete random variable takes at an exact value.	The PDF is the probability that a continuous random variable takes at a specified interval.
The PMF deals with the discrete random variables.	The PDF deals with the continuous random variables.
PMF is evaluated at specific point.	PDF is evaluated at specified interval
$f(x) = P(X = x)$	$P(x) = F'(x)$ where, $F(x)$ is CDF

Uses of Probability Mass Function

Probability Mass Function (PMF) is a fundamental concept in probability theory and statistics, particularly when dealing with discrete random variables. Some uses of the PMF are:

- **Describing Discrete Distributions:** PMF is used to define the distribution of discrete random variables such as the number of heads in a series of coin flips, the number of customers arriving at a store in a given hour, or the number of defective items in a batch.
- ***Calculating Probabilities:** PMF allows for the calculation of the probability that a random variable takes on a specific value.
- **Modeling and Simulations:** In simulation studies and probabilistic modeling, PMFs are used to generate random samples and to simulate scenarios involving discrete random variables.
- **Information Theory:*** PMFs play a crucial role in information theory, particularly in the calculation of entropy, which measures the uncertainty or randomness of a random variable.

***Example 1:** Probability mass function is given by: $f(x) = ax^2$ for $x = 0, 1, 2$ then, find the value of a.

***Solution:**

To find the value of a we use the PMF property.

$$\sum_{x \in S} f(x) = 1$$

$$f(0) + f(1) + f(2) = 1$$

$$a \times 0.2 + a \times 0.12 + a \times 0.22 = 1$$

$$a \times (0.2 + 0.12 + 0.22) = 1$$

$$a \times (0 + 1 + 4) = 1$$

$$5a = 1$$

$$a = 1/5$$

Example 2: From the below probability mass function table determine CDF.

X	1	2	3	4
P(X = x)	0.2	0.05	0.25	0.5

Solution:

To find the CDF for $P(X < 3)$ we use the property of PMF.

$$P(X \in E) = \sum_{x \in E} f(x)$$

$$P(X < 3) = P(X = 1) + P(X = 2)$$

$$P(X < 3) = 0.2 + 0.05$$

$$P(X < 3) = 0.25$$

So, CDF of $P(X < 3) = 0.25$

***Example 3:** Table below represents the PMF for a random variable X. Find the value of p.

X	2	4	6
P(X = x)	p	p + 0.5	0.2

Solution:

To find the value of p we use formula.

$$\sum_{x \in S} f(x) = 1$$

$$p + p + 0.5 + 0.2 = 1$$

$$2p + 0.7 = 1$$

$$2p = 1 - 0.7$$

$$2p = 0.3$$

$$p = 0.3 / 2$$

$$p = 0.15$$

***Example 4:** Find probability of good number of products = 8 if there 10 products and the probability of good product is 0.95.

Solution:

We will solve above question using Binomial distribution.

Here, $n = 10$, $x = 8$, $p = 0.95$

PMF of Binomial distribution is given by:

$$P(X = x) = {}^nC_x p^x (1 - p)^{n-x}$$

$$P(X = 8) = {}^{10}C_8 (0.95)^8 (1 - 0.95)^{10-8}$$

$$P(X = 8) = 45 \times 0.663 \times (0.05)^2$$

$$P(X = 8) = 29.835 \times 0.0025$$

$$P(X = 8) = 0.0746$$

$$\text{Probability of 8 good products} = 0.0746$$

Example 5: There are 10 pens, and the probability of defective pens is 0.1 then find the probability of 2 defective pens.

Solution:*

$$\text{Mean} = \lambda = np = 10 \times 0.1 = 1$$

$$P(X = x) = [\lambda^x e^{-\lambda}] / x!$$

$$P(X = 2) = [1^2 \times e^{-1}] / 2!$$

$$P(X = 2) = (0.3679) / 2$$

$$P(X = 2) = 0.184$$

$$\text{Probability of 2 defective pens is } 0.184$$